KAMPALA INTERNATIONAL UNIVERSITY-WESTERN CAMPUS
FUCULTY OF BIOMEDICAL SCIENCES
DEPARTMENT OF BIOCHEMISTRY
BMS 1.2 CAT

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Time: 1 hour 30 Minutes

ection A: Answer all questions in the MCQ answer sheet provided

1. $\Delta G°$ is defined as the
   a) Residual energy present in the reactants at equilibrium
   b) Residual energy present in the products at equilibrium
   c) Difference in the residual energy of reactants and products at equilibrium
   d) Energy required in converting one mole of reactants to one mole of products
2. What does the first law of thermodynamics state?
   a) Energy can neither be destroyed nor created
   b) Energy cannot be 100 percent efficiently transformed from one type to another
   c) All living organisms are composed of cells
   d) Input of heat energy increases the rate of movement of atoms and molecules
3. Which of the following statements is false?
   a) The reaction tends to go in the forward direction if $\Delta G$ is large and positive
   b) The reaction tends to move in the backward direction if $\Delta G$ is large and negative
   c) The system is at equilibrium if $\Delta G = 0$
   d) The reaction tends to move in the backward direction if $\Delta G$ is large and positive
4. The regulation of oxidative phosphorylation depends on
   a) Magnitude of ion motive force
   b) Magnitude of electron motive force
   c) Magnitude of proton motive force
   d) None
5. Anabolism and catabolism are chemically linked in the form of
   a) ADP
   b) ATP
   c) Phosphodiester linkage
   d) ASP
6. An endergonic reaction
   a) Proceeds spontaneously
   b) Does not require activation energy
   c) Releases energy
   d) Requires energy
7. Which out of the following is not a flavoprotein?
   a) Succinate dehydrogenase
   b) Cytochrome c

c) Xanthine oxidase

d) NADH dehydrogenase-Qreductase

8. Loss of electrons can be termed as

a) Metabolism

b) Anabolism

c) Oxidation

d) Reduction

9. The only membrane bound enzyme in the electron transport chain is

a) Succinate dehydrogenase

b) NADH dehydrogenase

c) ATP synthase

d) Acyl co-A dehydrogenase

10. Effect of valinomycin on oxidative phosphorylation does not involve

a) pH gradient across the inner mitochondrial membrane decreases

b) Rate of flow of electrons increases

c) Rate of oxygen consumption increases

d) Net yield of ATP decreases

11. Which one of the following respiratory chain components reacts directly with molecularoxygen?

A. Coenzyme Q

B. Cytochrome b

C. Cytochrome aa3

D. Cytochrome C1

12. Which of the following is a prosthetic group of NADH dehydrogenase complex?

A. FMN, Fe-S

B. FAD, Fe-S

C. Heme, Fe-S

D. PLP, Fe-S

13. Which of the following trace elements has role in ETC?

A. Iron

B. Iodine

C. Zinc

D. Fluoride

14. A naturally occurring uncoupler is:

A. Dicumarol

B. 2,4-DNP

C. Thermogenin

D. Salicylate

15. All of the following statements are true regarding ETC, *except*:

A. Located in inner mitochondrial membrane

B. Components are organized in decreasing order of redox potential

C. Involved with ATP synthesis

D. Cyanide inhibits electron flow

16. In oxidative phosphorylation, the oxidation of one molecule of NADH producesapproximately:

A. 3.5 ATP molecules

B. 2.5 ATP molecules

C. 4 ATP molecules

D. 1.5 ATP molecule

Which of the five enzyme complexes in the electron transport system is directly involved in ATP synthesis:

A. Complex I
B. Complex II
C. Complex IV
D. Complex V

8. Individuals with disorders of the respiratory chain are often placed on supplements containing riboflavin and coenzyme Q. Whichof the following if the role of coenzyme Q in the respiratory chain?
   A. It links flavoproteins to cytochrome b, the cytochrome of lowest redox potential
   B. It links NAD-dependent dehydrogenases to cytochrome b
   C. It links each of the cytochromes in the respiratory chain to one another
   D. It is the first step in the respiratory chain

19. Electron transport and phosphorylationcan be uncoupled by compounds thatincrease the permeability of the innermitochondrial membrane to
   A. Electrons
   B. Protons
   C. Uncouplers
   D. All of these

20. An uncoupler of oxidative phosphorylationsuch as dinitrophenol
   A. Inhibits electron transport and ATP synthesis → oligomycin
   B. Allow electron transport to proceed withoutATP synthesis
   C. Inhibits electron transport without impairmentof ATP synthesis
   D. Specially inhibits cytochrome b

21. How many ATP molecules can be derived from each molecule of acetyl CoA that enters the Kreb's cycle?
   a. 6
   b. 12
   c. 18
   d. 38

22. Which of the following reactions generates ATP?
   a. Glucose to glucose-6-phosphate
   b. Pyruvate to lactate
   c. PEP to pyruvate
   d. Glucose-6-phosphate to fructose-6-phospahate

23. Glycogenin is.....................
   A. Uncoupler of oxidative phosphorylation.
   B. Polymer of glycogen molecules.
   C. Protein primer for glycogen synthesis.
   D. Intermediate in glycogen breakdown.

24. All are intermediates of glycolysis except,
   a. Glucose-6-phosphate
   b. Fructose-1, 6-bisphosphate
   c. Fructose-6-phosphate
   d. Glycerol-3-phosphate

25. The first substance produced in the citric acid cycle is:
   a. Acetyl CoA

b. Oxaloacetate
c. ATP
d. Citrate

26. The end product(s) of anaerobic catabolism of glucose in muscle tissue is (are):
a. Carbon dioxide and water
b. Lactate.
c. Sucrose.
d. Ethanol

27. If the glucose level in the body is high, the body produces more:
a. Insulin
b. Thiamine.
c. Glucagon.
d. Epinephrine

28. All of the following compounds are intermediates of TCA cycle except:
a. Malate
b. Pyruvate
c. Oxaloacetate
d. Fumarate

29. MacArdle's disease involves a deficiency of which enzyme?
a. Acid maltase
b. Glucose-6-phosphatase
c. Hepatic phosphorylase
d. Muscle phosphorylase

30. The following metabolic abnormalities occur in Diabetes mellitus, except:
a. Increase plasma free fatty acids
b. Increased pyruvate carboxylase activity
c. Decrease lipogenesis
d. Decrease gluconeogenesis

31. A substance that is not an intermediate in the formation of D-glucuronic acid from glucose is:
a. Glucose-1-phosphate
b. 6-Phosphogluconate
c. Glucose-6-phosphate
d. UDP–Glucose

32. The glycolysis is regulated by?
A. Hexokinase.
B. Phosphofructokinase
C. Pyruvate kinase.
D. All of the above.

33. Which of the following statements regarding TCA cycle is true?
A. It is an anaerobic process.
B. It occurs in cytosol.
C. It contains no intermediates for Gluconeogenesis.
D. It is amphibolic in nature.

34. Pyruvate, the end product of glycolysis, can acquire an amino group by transamination, producing the amino acid

a. Glycine
b. Alanine
c. Leucine
d. Arginine

35. Production of citrate from oxaloacetate and acetyl CoA is catalyzed by
a) Citrate synthase
b) Citrate synthatase
c) ATP citrate lyase
d) None of the above

36. The anaerobic conversion of glucose to pyruvate is called:
a) Glycolysis.
b) Embden-Myerhof pathway.
c) Fermentation.
d) Anaerobic

37. Which of the following statements is correct?
a. Anaerobically, oxidative decarboxylation of pyruvate forms acetate that enters the TCA cycle
b. In anaerobic muscle, pyruvate is converted to lactate
c. Reduction of pyruvate to lactate generates a coenzyme essential for glycolysis
d. Under anaerobic conditions, pyruvate does not form because glycolysis does not occur.

38. One of these is not correct about glycogenolysis is
(a) It can take place after a protein meal glycogen → glucose
(b) It can take place during emotional stress
(c) It can take place during strenuous exercise
(d) It can take place during fasting

39. Which of these pairs contain a hormone that cannot trigger glycogenolysis
(a) Catecholamines and glucagon
(b) Insulin and catecholamines
(c) Glucagon and thyroid hormone
(d) Thyroid hormone and catecholamines

40. Gluconeogenesis is vital in glucose metabolism for the following reasons except ........
(a) To keep basal level of blood glucose needed as source of energy for special tissues
(b) To maintain cellular level of intermediates of TCA cycle in the cell
(c) Glucose is the only available fuel for skeletal muscles in anaerobic conditions
(d) None of the above

41. Which of the following will be found in the core of a lipoprotein?
a) Phospholipids, because they are hydrophobic
b) TGs, because they are hydrophilic
c) Cholesterol and cholesterol esters, because they are amphipathic
d) TGs and cholesteryl esters, because they are hydrophobic

42. Which of the following statements is true of fatty acids in living cells?
a) synthesis takes place in the mitochondria, β-oxidation takes place in the cytoplasm
b) both synthesis and β-oxiadation take place in the cytoplasm
c) both synthesis and β-oxiadation take place in the mitochondria
d) synthesis takes place in the cytoplasm, β-oxidation takes place in the mitochondria

43. The following enzymes are involved in transportation of fatty acids into mitochondria except.........
a) Acyl CoA Synthetase
b) Acyl CoA Synthase acetyl CoA + OAA → citrate
c) Carnitine Acyl transferase I
d) Carnitine Acyl transferase II

44. A fatty acid molecule is degraded to yield ATP in the order of the following metabolic pathways

a) TCA cycle, electron transport, β-oxidation
b) TCA cycle, β-oxidation, electron transport chain
c) β-oxidation, electron transport chain, TCA cycle
d) β-oxidation, TCA cycle, electron transport chain

45. In the β-oxidation pathway, the four types of reactions occur in the following order oxidation → hydration → oxidation → cleav
   a) dehydrogenation ,hydration, dehydrogenation, thiolysis
   b) thiolysis, dehydrogenation, hydration, dehydrogenation
   c) dehydrogenation, hydration, thiolysis, dehydrogenation
   d) hydration, thiolysis, dehydrogenation, dehydrogenation

46. Which of these statements is true of β-oxidation of odd carbon fatty acids?
   a) propionyl CoA and acetyl CoA are converted to succinyl CoA in TCA cycle
   b) succinylCoA and acetylCoA are the end products
   c) both propionyl CoA and acetyl CoA can react with oxaloacetate to form citrate
   d) propionylCoA enters TCA cycle after its conversion to succinyl CoA propionyl CoA → succinyl CoA

47. Which of these statements does not occur in ketone bodies formation?
   a) Accumulation of acetyl CoA leads to ketone bodies formation
   b) β-hydroxybutyrate is formed by reduction of acetoacetate
   c) Ketogenesis occurs after excess β-oxidation
   d) Ketone group in acetoacetate and β-hydroxybutyrate raises blood pH beyond 7.4

48. In fatty acid synthesis, which of the following biochemical reactions do not occur in the process of adding t carbon atoms.
   a) Condensation of malonyl group with acetyl group and consequent decarboxylation
   b) Supply of H atoms by NADPH to reduce the formed keto group
   c) Supply of H atoms by NADH to reduce the formed keto group
   d) Supply of H atoms by NADPH to reduce the double bond

49. Which of these is false about atherosclerosis?
   a) It is due to imbalance between demand and supply of cholesterol to tissues
   b) It occurs as a result of accumulation of cholesterol in blood vessels.
   c) It occurs when there is high level of LDL-bound cholesterol
   d) It occurs when there is high level of HDL-bound cholesterol

50. ……… and ………are regulatory enzymes in fatty acid synthesis and beta oxidation respectively.
   a) acetyl CoA carboxylase and ACP
   b) acetyl CoA carboxylase and carnitine acyl transferase I
   c) ACP and fatty acid synthase
   d) carnitine acyl transferase I and ACP

51. Blood rich in glucose after a carbohydrate meal can trigger the following except…………
   a) synthesis of fatty acids
   b) beta oxidation of fatty acids
   c) synthesis of triacyglycerols
   d) synthesis of cholesterol

52. Which of the following is not true of eicosanoids ?
a) They are derived from arachidonate
b) They act as local hormones which trigger inflammatory response
c) Some are constrictors while others are dilators of blood vessels.
d) They can be classified as prostaglandins, leukotrienes and thromboxanes

53. Which of the following statements is not true of the enzyme cyclooxygenase?
a) It can be inhibited by aspirin and ibuprofen
b) It is otherwise called prostaglandin H2 synthase
c) It is also referred to as lipoxygenase because it is known for oxidizing lipids.
d) It is found in the pathway of prostaglandin synthesis

54. A round of β oxidation of fatty acids yields the following excpt...
a) NADH
b) NADPH
c) $FADH_2$
d) Acetyl CoA

55. Which of these statements is false?
a) LDL carries bad cholesterol when the latter is excess for the need of extrahepatic tissues
b) LDL carries good cholesterol when the latter is excess for the need of extrahepatic tissues
c) HDL clears excess endogenous and exogenous cholesterol to the liver for bile acid synthesis
d) HDL cholesterol content can be regarded as good cholesterol

56. The following order of biomolecules represents stages in cholesterol biosynthesis
a) acetyl CoA, squalene, mevalonate, isoprene
b) acetyl CoA, mevalonate, isoprene, squalene
c) acetyl CoA, isoprene, mevalonate, squalene
d) acetyl CoA, mevalonate, squalene, isoprene

57. The following are lipotropic factors except:
a) Choline
b) Methionine
c) Tyrosine
d) Beteine

58. Cholesterol is a vital biomolecule in the synthesis of the following pairs except ..........
a) Bile acids and Progesterone
b) ATP and Acetyl CoA
c) Sterols and Cholic acid
d) Testosterone and vitamin D

59. Niemann-Pick disease is due to deficiency of the following enzyme
a) Hexosaminidase A
b) Sphingomyelinase
c) Ceramidase
d) β-Glucosidase

60. Ketosis is associated with the disease:
a) Nephritis
b) Diabetes mellitus
c) Edema
d) Coronary artery diseases

**Section B: Fill in the blanks in the answer sheet provided**

3'5' cyclic AMP

1. Examples of high energy compounds include ...PEP, Pyrophosphate...... and ....................................
2. Mechanisms of coupling include .Obligatory..common....intermediate..........and .....Through..compound..of..high..energy..compound.
3. State two antidotes for cyanide poisoning of the respiratory chain:...Sodium...thiosulphate....... and ......sodium..nitrate..........
4. Cytosolic NADH is transported into the mitochondria for oxidation by ....Malate-aspartate shuttle.

5. Name two sources of reducing equivalents that are oxidized in the respiratory chain

.............................................and.............................................

6. List two major sources of reactive oxygen species in the cell:...Peroxisomes.....And

......Mitochondria...................... & DNA, RNA.

7. Oxygen radical reactions occur with ............................................. .....and

.....................................cellular components

8. A phenomenon where some organisms extract chemical energy and convert it to light is known as .............................................

9. Bongkrekic acid is a respiratory toxin implicated in deaths resulting from eating coconut based product known as tempe in Indonesia. It is produced in fermented coconut contaminated by the bacterium Bukholderia gladioli. It exerts its action by inhibition of

......Adenine Nucleotide translocase.

10. The component of electron transport chain that is not involved in pumping electrons into the inter-membrane space is .....................................

11. Krebs cycle and the electron transport chain occur in the.........................

12. An essential enzyme for converting glucose to glycogen in liver is

.............................

13. Which of the following is a substrate for aldolase activity in glycolytic pathway?

14. Important hormones involved in carbohydrate metabolism is .........................

15. Dehydrogenases involved in HMP shunt are specific for .............................

16. The fructokinase reaction in fructose metabolism produces ...........................

17. TCA cycle occurs in .................

18. Which major process produces blood glucose after 2 days of fasting?

19. The glucose transporters responsible for the transport of glucose into the muscle cell are?

20. In which part of the cell does glycolysis take place?

21. Statin drugs inhibit the enzyme ..................................... of cholesterol biosynthesis

22. ..............................................is a drug that increases the rate of LDL metabolism and may block the intestinal transport of cholesterol

23. The only way of elimination of cholesterol from the body is by converting it to ..............

24. Tangier disease is due to the deficiency of....HDL deficiency.........lipoprotein

25. The lipoprotein with the highest percentage concentration of TAGs is the ..........................

26. The lipoprotein with the highest percentage concentration of proteins is the

.............................

27. The lipoprotein with the lowest density is the ..............................................

28. The lipoxygenase pathway is used in the synthesis of this eicosanoid: ............................

29. Non steroidal anti inflammatory drugs (NSAIDs) such as aspirin inhibit the enzyme

.............................. of eicosaniod synthesis.

30. The ketone body that is volatile and breathed out, is ..........................................

donor.

NADH / ~~FADH~~ → strong ē donor